Jupiter Suite Pro

Web Scraper · Media Download · NLP Data Prep

Full Features & Product Description

Version 5.36 family · Jupiter International Group

A desktop suite for capturing social and web content, downloading media with platform-aware fallbacks, and optional NLP data-preparation / modeling apps that share the same licensing mesh. Capture runs through a real browser session to help pass common Cloudflare and anti-bot checks.

1. Overview

Jupiter Suite Pro is a Windows desktop application built around a Chrome extension bridge and a native host. The operator opens pages in the browser, starts capture from the extension, and the suite receives structured DOM/HTML, reviews/comments, media candidates, and session cookies. The main app parses the capture, queues jobs, exports datasets, and downloads video/audio through yt-dlp, progressive CDN URLs, or browser-assisted paths.

The product is designed for research, archiving, content operations, and NLP pipelines: one capture flow feeds reviews, media, and debug artifacts. Because capture uses the live browser profile, sites protected by Cloudflare or similar anti-bot layers can often be reached the same way a normal user would.

2. Architecture (high level)

- **Chrome extension** — capture UI, progress/ETA, platform-specific open-comments and expand-replies, stream sniffer, cookie read, optional audio-language meta for YouTube.
- **Native messaging host** — JupiterMediaHost.exe bridges extension and desktop app.
- **Capture server** — receives URL, HTML, title, cookies; hands off to scraper/parser off the UI thread.
- **Scraper + media panels** — jobs queue, extracted data, media download (yt-dlp / CDN / browser).
- **Cloudflare / anti-bot path** — capture through the real Chrome session (cookies, TLS fingerprint, solved challenges) instead of headless-only bots.
- **Licensing / VM licensing** — optional hardware-locked activation shared with companion Data Prep apps.
- **Build path** — PyInstaller or Nuitka onefile/standalone, optional safe obfuscation, VM files, xollvm where supported.

3. Web Scraper features

3.1 Capture & jobs

- Capture this page from the extension with live progress bar, percentage, STOP, and optional extension sound toggle.
- Tiered ETA (S/M/L/XL) and fast-stop when scroll stalls or comment growth stabilizes.
- Job queue with multi-select remove; optional update of SiteProfile history when removing URLs.
- Extracted Data panel for reviews and fields; export JSON/CSV-style datasets.
- Save Debug Files / webdebug element and element_html dumps for analysis.
- Processing history logs for completed sessions where enabled.

3.2 Bypass Cloudflare and anti-bot checks

Jupiter is designed to work with the operator's real browser session rather than a detached headless scraper:

- **Live Chrome profile** — same cookies, local storage, and login state the user already has.
- **Human-like page context** — DOM is read after the page has loaded in a normal browser, including after Cloudflare or similar interstitials have been passed interactively.
- **Session cookies to the app** — extension can pass cookie headers for media download (e.g. yt-dlp cookiefile) so authenticated or challenge-cleared sessions carry through.
- **Soft refresh / stay-on-URL flows** — platform sequences can refresh or re-open panels without abandoning the cleared session.
- **Practical limit** — automated solving of every CAPTCHA or enterprise bot score is not guaranteed; success depends on the live session and site policy.

3.3 Platform-aware social capture

Major networks use **separate** capture sequences (not one shared expand/CDN path) to avoid cross-site regressions:

• **Facebook** — comments expand, view-more loops, video fallback via API/CDN ranking, audio-aware media selection.

• **Instagram** — /p/ vs /reels/ handling, comments panel, view replies, popup-aware extension behavior.

• **TikTok** — comments balloon, Comments vs You May Like, view N replies with multi-click expand, stay on video URL, CDN video path.

• **Douyin** — open comments panel, expand replies, refresh-before-capture option, media CDN fallback.

• **Bilibili** — individual path, optional soft refresh for cookies, noise filter for auto-caught unrelated media.

• **LinkedIn** — popup comments, scroll in panel, media/CDN where available.

• **YouTube** — watch and Shorts, comments panel open, reply expanders, individual media path via yt-dlp + HTML progressive + sniffer; audio language from live player meta.

• **X (Twitter)** — show probable spam / expand paths where supported.

• **Reddit** — more replies / view more comments cycles with refresh-aware open behavior.

3.4 Media download

• Media Download panel: video / audio / image queues, quality presets, optional browser cookie source for yt-dlp.

• Primary path yt-dlp; fallbacks: HTML progressive (streamingData videoId match), extension sniffer / jupiter-stream, social CDN progressive.

• API CDN direct download logging and progress aligned to item counts where used.

• YouTube: prefer watch?v= URL; player-aligned audio language via extension meta tags; filename suffix by language actually downloaded.

• Optional Node.js portable + EJS/challenge support for yt-dlp n-parameter where the environment allows.

• Bundled or detected ffmpeg for merge/remux.

4. Chrome extension UX

- Capturing progress bar with cyan glow and percentage beside STOP (red glow).
- Sound ON/OFF under STOP for capture progress cues.
- Revive rules: after 100% or STOP with data sent, disable revive; enable when a new capture starts.
- Popup / SPA awareness so capture runs in the visible context.
- YouTube audio: list tracks from the live player and inject jupiter-audio-lang / jupiter-audio-tracks meta for the desktop app.

5. Licensing, protection & builds

- Licensed vs unlicensed builds (activation dialog, hardware lock when enabled).
- Optional VM licensing files (licensing_vm.py + vm_secret.key), generatable when missing.
- safe_obfuscate / AST pre-Nuitka obfuscation for sensitive modules (avoid PyArmor+Nuitka conflict).
- Nuitka whole-app onefile or standalone; dual Python 3.12 / 3.14 venv options; MSVC / MinGW / Zig backends where configured.
- Optional xollvm (prebuilt or from source) with experimental flags; large disk/time cost.
- Optional post-EXE light packer (LZMA + stub) for PE hardening experiments.
- Core protection focus: licensing, capture_server, stream_downloader, media_types, chrome_cookie_unlock.

6. Companion NLP apps (shared UI suite)

Four related executables share UI patterns and can share licensing with the web scraper:

- **Classical (Data Preparation)** — dataset load, prep, charts, spaCy components, optional ipynb export with engine-selected modeling steps.
- **Modeling** — consumes latest data-prep manifests/reports; combined HTML report chain; transformers/torch optional.
- **SEA-Lion** — region/language-oriented prep; detect Classical and/or Web Scraper datasets.
- **SEA-Lion Modeling** — manifests from SEA-Lion and/or data-prep; API-oriented notebook generation options.

Shared features include boot screen/logo/SFX, backup folders next to the EXE with date/time stamps (backup_dataset, export_dataset, export_debug_log), Settings for backup location, and optional ipynb with stage charts.

7. Product UX highlights

- Boot logo, ring, progress bar, and power-up / success / error SFX (when assets are bundled correctly).
- Non-blocking parse of large DOMs on a worker thread.
- Console-off shipping builds; diagnostic console option during development.
- Discoverable mesh ports for multi-app connectivity (scraper, media host, peers).

8. Known limits (transparent)

- Site UIs change often; expand-reply and anti-bot behavior may require profile or extension updates.
- Cloudflare / bot protection: relies on a real, cleared browser session — not a guarantee against every CAPTCHA or enterprise rule set.
- YouTube AI-dub / multi-audio depends on what yt-dlp can list (cookies, n-challenge/EJS, SABR).
- Browser cookie DB may be locked while Chrome is open; extension cookie headers / cookiefile are preferred for some sites.
- Heavy Nuitka + xollvm builds need substantial disk, RAM, and flag tuning on real Windows machines.

9. Feature map (summary)

Area	Capabilities
Capture	Extension progress, STOP, ETA, multi-platform comments/media
Anti-bot	Live Chrome session; Cloudflare / challenge-cleared cookies to app
Media	yt-dlp, CDN progressive, sniffer, ffmpeg, YT language meta
Data	Exports, webdebug, stamped backups, NLP companion apps
Security	License/VM, obfuscation options, Nuitka/PE experiments